

# Hearing Loss Is Reflected in Conversational Pupil Responses

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## Abstract

Pupillometry has been shown to be a correlate for expected mental workload in a variety of cognitively demanding tasks, including those related to listening. Recent work has extended the application of pupillometry to interactive conversation by measuring how the pupil responds around turn-taking. This study furthers the application of pupillometry to conversation by evaluating differences in pupil responses during conversations taking place between individuals with and without hearing loss. Pairs of older Danish talkers (in which one had hearing loss, and one did not) held conversations in a variety of acoustic conditions, and their pupil responses were recorded throughout. Our findings show that both background noise and hearing loss modulate pupil responses during conversation, and further show correspondence between pupil dilation responses and measures of turn-taking behavior which have been suggested to be indicative of increased difficulty.

## Keywords

pupillometry, turn-taking, hearing loss, listening effort, conversational dynamics, temporal response function

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## Introduction

Hearing loss profoundly impacts one's ability to communicate easily and effectively, potentially leading to social withdrawal and feelings of isolation and depression (Arlinger, 2003). Typical evaluations of hearing loss often only consider how well a person can hear, in isolation, for example with pure tone audiometry or speech-in-noise tests (Billings et al., 2024; International Organization for Standardization, 2010). However, one potential reason why communication becomes so difficult for talkers with hearing loss (HL) is that conversation involves more than just hearing and listening.

Previous psycholinguistic studies have found that, while a typical floor-transfer offset (FTO, i.e., the amount of time between subsequent turns) in conversation is on the order of 200 ms, it takes upwards of 600 ms to plan and begin producing a response (Levinson & Torreira, 2015). This suggests that talkers must be planning what they'll say next while still listening to their partner's speech. Given this observation, it is possible that HL talkers experience difficulty not only in listening, but also in performing the concurrent processing necessary to maintain fluid communication.

Some prior work in this area has taken a behavioral approach to investigating how difficulties such as hearing loss, background noise, and second language affect the timing of conversational turn-taking (Petersen et al., 2022; Sørensen et al., 2021, 2024). For example, both talkers with typical hearing (TH) and HL talkers exhibit longer and more variable FTOs when

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communicating in the presence of background noise (Sørensen et al., 2024). It has been suggested that these changes in the FTO may result from increased listening demands leaving less resources available for simultaneous speech planning, which in turn delays the onset of the listener's next turn. Perceptual studies have also demonstrated that the timing of turn starts influences the subjective impressions of conversations, with longer FTOs resulting in conversations being rated as having poorer flow and lower connection between interlocutors, and being more difficult to follow (Petersen, 2025; Templeton et al., 2022). Therefore, evidence suggests that in addition to affecting speech understanding, listening difficulty can influence the timing of speech interactions with consequent effects on how the conversation is perceived by the interlocutors.

This suggests that there are insights about hearing loss and its effects that can be gained by studying communication. One potential way in which the effects of hearing loss on communication could be further evaluated is to characterize the amount of cognitive effort that talkers exert while communicating, which would enable the study of when increased effort occurs and why. One recent focus of research in hearing science has been to evaluate listening effort, defined as 'the deliberate allocation of mental resources ... when carrying out a listening task' (Pichora-Fuller et al., 2016), to better understand hearing loss. A commonly used metric to assess both listening effort and cognitive effort, more generally, is the pupil dilation response, which has been shown to be a correlate for expected mental workload under a wide range of cognitively demanding tasks (Beatty, 1982; Kahneman & Beatty, 1966), including some related to listening (Fiedler et al., 2023; Koelewijn et al., 2014; Wendt et al., 2018).

Listening effort may not fully capture the cognitive demands of conversation, though, as it is expected that a substantial amount of effort must be devoted both to speech planning, and to the production of speech itself. Some prior work has evaluated pupil responses in the context of speech production, though, to investigate the time-course of effort related to speech planning (Barthel & Sauppe, 2019; Paulus, 2021; Sauppe, 2017). These works have found that a dilation of the pupil is induced in a manner time-locked to when speech planning can begin, and it's suggested that this is indicative of the cognitive effort required for planning.

Although studies have separately evaluated effort via pupil responses around listening and speaking, the fast-paced demands of conversation necessitate that these processes must often happen in parallel (Levinson & Torreira, 2015). Therefore, it is not clear the extent to which the findings from either the listening-only or production-only side will generalize to interactive communication. Further, it can be challenging to infer effort during interactive communication given its dynamic, spontaneous nature. For example, while typical pupillometry experiments would tightly control the visual field to diminish the effects of the pupillary light reflex or the near pupil response (Mathôt, 2018; Winn et al., 2018), it is infeasible to institute such controls in a manner which is representative of natural conversation. There have been recent attempts to apply pupillometry to conversation, though, first by comparing descriptive statistics between tasks of different expected difficulty levels (Li et al., 2020), and then using a deconvolution approach to assess whether the pupil responds consistently around turn-taking (Masters et al., 2025). These studies found that the pupil is sensitive to both task and background noise induced difficulties, and that despite the lack of visual field controls, it is possible to measure momentary changes in the pupil size during communication.

This work aims to extend the application of pupillometry to conversation by assessing whether differences in pupil responses during conversation are observed between TH and HL talkers. We first evaluate pupil responses over entire conversations and then further analyze pupil responses for systematic changes that occur around turn-taking. We expect, consistent with previous findings from listening experiments (e.g., Zekveld et al., 2023), that background noise will induce larger pupil sizes overall, and that the pupil responses of HL talkers will exhibit larger effects of fatigue, such as a smaller average pupil size and a steeper drift in pupil size over the course of a conversation, due to increased listening and communication difficulty. We further expect that HL talkers will exhibit significantly different pupil responses to turn-taking due to increased processing load for speech comprehension. We also assess turn-taking behavior to determine if differences in turn lengths or FTOs partially contribute to any effects observed on the pupil response to turn-taking.

## Methods and Materials

### *Participants and Experimental Design*

This study involves the re-analysis of data that were collected in a different study by researchers at Eriksholm Research Center who are not involved in the current investigation. The purpose of that study was to explore the potential of a novel proprietary noise reduction algorithm. Thus, the remainder of this subsection briefly describes the participants recruited and methods used in the previous study.

In total, 48 older Danish participants were recruited. Half of the participants had age-adjusted typical hearing as per the International Organization for Standardization's ISO-7029 (ages  $65.9 \pm 6.5$ ). The other half had differing degrees of age-related sensorineural hearing loss (HL) (ages  $72.3 \pm 8.4$ , loss profiles: 7 mild, 8 mild-to-moderate, 4 moderate, and 5 severe).

Hearing status was evaluated using cross-ear average pure-tone audiometry at octave frequencies from 500 to 4k Hz. An equal number of male and female participants with and without hearing loss were recruited. All participants were screened for cognitive impairment using the Montreal Cognitive Assessment test (Nasreddine et al., 2014), any potential participants that scored less than 23 were deemed as having mild or greater cognitive impairment and thus excluded from the study.

Gender-mixed pairs were constructed in which one participant had hearing loss, and one did not. Each pair was seated face-to-face in a sound-treated room and held a series of conversations under different acoustic conditions. Conversations took place either in quiet (Q) or in the presence of background noise, at levels of 60 dBA (N60) and 70 dBA (N70), that was played back over a horizontal, circular, loudspeaker array positioned at approximately head height. The HL talkers wore hearing aids set with unity gain (i.e., no amplification) and received one of three levels of noise reduction: none, low, or high. A single conversation was held in all combinations of background noise and noise reduction settings for each pair, yielding 9 conditions and conversations. The order of the conditions was counter balanced. Speech from participants was recorded using directional head-worn microphones. Talkers also wore Tobii Pro 3 glasses, which measured pupil size and eye-gaze position with a 50 Hz sample rate, as metrics related to eye-gaze were intended to be explored in the previous study.

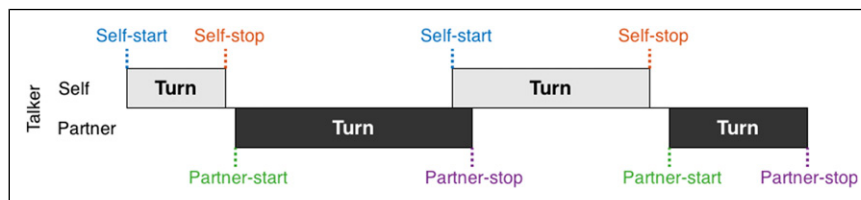
Conversations were elicited using the DiapixUK collaborative spot-the-difference task (Baker & Hazan, 2011), but with DiapixDK pictures (Sørensen, 2021), which include Danish rather than English signage. In this task, each participant in a pair was given an image, which differed only slightly from an image that was given to their partner (e.g., a red car in one's image versus a blue car in the partner's). Talkers are then tasked with describing what they see, and identifying what differences exist between their own image and their partner's, while only seeing their own image. Each pair of pictures contained 12 differences. Pairs were given up to 4 minutes to find all 12 differences in each of the Diapix images. After either 4 minutes had passed or all differences were found, whichever came sooner, the operator instructed the pair to advance to the next image. Across the experiment, most conversations reached the 4-minute limit before being stopped, with only ~7% of conversations stopped early due to all differences being found in a given pair of pictures. There was no other interference from the operator during the conversations.

In the present study, we evaluate only hearing status and background noise as sources of difficulty and therefore collapse all analyses across noise reduction settings.

### Speech Preprocessing and Turn-Taking Identification

Voice activity detection (VAD) was performed on the headset microphone signals using individually defined energy thresholds in 5 ms windows with 1 ms of overlap. The VAD signals were bridged to remove short acoustic bursts (<90 ms), which were assumed to be non-speech, and short silences (<180 ms), which were assumed to be associated with the production of stop consonants. These durations are defined based on the expected minimum duration of voiced speech, and the maximum typical duration of stop-consonants, respectively, as suggested by Heldner and Edlund (2010).

From the VAD signals, turn-taking was identified to have occurred when one talker had been speaking for at least 90 ms while their partner had been silent for at least 180 ms. These durations are selected for the same reasons for which they were used in the VAD method. The start and end of conversational turns were then identified as the onset and offset of speech for each occurrence of turn-taking. FTOs were computed as the amount of time between the end of one talker's turn and the start of the other talker's subsequent turn. Turn durations were computed as the amount of time between the onset and offset of the same turn. As we aimed to measure the pupil responses to turn-taking, the turn onsets and offsets were further segmented as belonging either to oneself (i.e., talking) or to one's partner (i.e., listening). This leads to the identification of four types of turn-taking events of interest in conversation: self-start and self-stop correspond to the onset and offset of speech production, and partner-start and partner-stop correspond to the onset and offset of a partner's speech production. These four types of turn-taking events are denoted on a sample dialogue exchange in Figure 1. Logical time-series were constructed for each of the four types of turn-taking events. These time-series had a value of 1 at samples where the corresponding event occurred and a value of 0 everywhere else.



**Figure 1.** The four conversational turn-taking events of interest denoted on a sample dialogue exchange between two talkers

### Pupil Response Preprocessing

For each conversation, pupil data were first de-blinked by identifying samples with pupil sizes greater than 3 standard deviations from the mean and removing fixed windows of 50 ms before to 150 ms after each of these artifacts. From each talker in each conversation, the eye with the least missing data was used for further analysis. If more than 40% of samples were missing in both eyes for a given talker in a conversation, that talker's pupil responses in that conversation were excluded from analysis. This process resulted in the exclusion of 74 individual pupil responses (~17% of all recordings, breakdown: 12 TH in Q, 14 TH in N60, 13 TH in N70, 12 HL in Q, 12 HL in N60, and 11 HL in N70). Within a conversation, missing data were interpolated using a cubic spline.

For analyzing pupil responses to turn-taking, the pupil data were further filtered using a 5<sup>th</sup> order robust detrending method, which enables slow drifts in signals to be removed while minimizing the influence of any measurement artifacts which typical detrending methods (e.g., high-pass filters) are prone to (De Cheveigné & Arzounian, 2018) to diminish the effects of arousal-induced pupil dilations at the onset of conversations. For the conversation level analysis of pupil size, this detrending was not performed so as to maintain the absolute size of the pupil. In both cases, the data were then zero-phase lowpass filtered using a linear phase finite impulse response filter with a cut-off frequency of 10 Hz and subsequently down sampled to 20 Hz, as changes in pupil size are uncorrelated across the eyes above this rate (Winn et al., 2018) and thus less likely to be indicative of higher-level processing. For analyzing the responses to turn-taking, pupil data were further standardized within talkers such that, for each talker, the set of pupil responses across all conversations had zero-mean and unit variance.

### Extracting Pupil Responses to Conversational Turn-Taking

For each conversation, multi-input single-output temporal response functions (TRF) were computed using the mTRF toolbox (Crosse et al., 2016) with the response signal set to the preprocessed pupil data and the stimulus signals as the set of time-series corresponding to the occurrences of each of the four turn-taking events. The duration of the TRFs were specified *a priori* at 2.5 seconds forward and backward from each event (i.e., total duration is 5 seconds). This duration was selected based on the expectation that it would contain the entirety of most turns, and thus that pupil responses invoked by these turns would not be truncated by the chosen window size. In this study, ~90.5% of all turns were less than 2.5 seconds, and this proportion was relatively constant across conditions (~90.2% in Quiet, ~89.9% in N60, 91.5% in N70). The regularization parameter was optimized to minimize the average reconstruction error across all conversations and participants. A two-stage approach was taken to select this regularization parameter. The first stage consisted of a coarse-search over possible regularization values using 20 log<sub>10</sub>-spaced steps between 10<sup>-2</sup> and 10<sup>2</sup>. Each regularization value was evaluated using 10-fold cross validation, and the value with the lowest average reconstruction error ( $\lambda_1$ ) was passed to the second stage. A finer search was then conducted over 20 linearly spaced steps ranging from  $\lambda_1/5$  to  $5\lambda_1$ , again using 10-fold cross validation, and the regularization value which minimized the average reconstruction error in this stage was then used in deriving the TRFs presented in this study (resulting  $\lambda = 5.22$ ). The outcome of this process was a set of four response curves (i.e., TRFs) corresponding to each of the four turn-taking events that indicated how the pupil responded consistently around each event, for each talker in each conversation. Two further eye-gaze based covariates were included as input features in the TRF models based on the experimental setup. The first indicated points in time when a talker looked from their Diapix image to their partner, and the second indicated when a talker looked from their partner to their image. These moments were identified as when the depth of a talker's gaze, as measured by the Tobii gaze position estimates, crossed the midpoint of the distance between the image and their partner. Both features were structured similarly to the turn-taking features (i.e., 1 when the event occurred, 0 at all other points in time). These additional features were included only to control for the influence of eye-gaze and accommodation on the pupil dilation response (e.g., Masters et al., 2025) and thus will not be presented in this study.

### Statistical Methods

Turn-taking behavior was analyzed using generalized linear mixed effects models (GLMM, fitted with *glmer()* from the *lme4* package in R). The effects of noise and hearing loss on FTOs were assessed using a model of the form 'FTO ~ Noise\*HearingStatus + Pre Floor-Transfer (FT) Interpausal Unit (IPU) Duration + Post FT IPU Duration + (1 | Talker) + (1 | Pair)', where noise and hearing status correspond to the level of background noise and whether a talker had TH or HL, and talker/pair are unique identifiers for each participant and pair, respectively. The pre/post FT IPU Duration factors indicate the durations of the IPU (i.e., utterances) immediately preceding and following the floor-transfer, as these have been shown to affect FTO duration (Roberts et al., 2015). The continuous duration predictors were scaled prior to fitting. The FTO GLMM was fitted using a gamma distribution with a log link function. Given that gamma distributions only contain non-negative

values, the FTO distribution had to first be right-shifted and/or truncated. To do this, we estimated the mean of a gamma distribution with the same variance and skewness as the distribution of all FTOs. The FTO distribution was then right-shifted by the difference between this estimated gamma mean and the true mean of all FTOs. FTOs which remained negative after this shift were excluded from analysis (resulting shift: 157.3 ms, 9.3% of FTOs excluded). The total counts of FTOs excluded were relatively balanced across conditions (289 TH in Q, 296 TH in N60, 306 TH in N70, 320 HL in Q, 293 HL in N60, 252 HL in N70). This procedure aligns with that used in previous studies of turn-taking behavior (Ellag et al., 2025; Masters & MacDonald, 2025; Sørensen et al., 2024).

The durations of turns were analyzed using a similarly fitted gamma distribution based GLMM with a log link function. Given that the minimum turn duration is 90 ms, defined by the threshold criteria for turn-taking to occur, the turn duration distributions were first left-shifted by 90 ms. A GLMM of the form ‘Turn Duration ~ Noise\*HearingStatus + (1 | Talker) + (1 | Pair)’ was then fitted to the durations of all turns across all conversations.

Conversation level analysis of pupillometry data was performed using linear mixed effects models (LMM, fitted with the *lmer()* function from the *lme4* package in R) of the form ‘Feature ~ Noise\*Hearing Status + (1 | Talker) + (1 | Pair),’ where ‘Feature’ represented the parameter which was being analyzed in each model. The specific features analyzed were the mean pupil diameter for each talker in each conversation, and the drift of pupil size from its baseline state. To quantify the drift from baseline over the course of a conversation, we compute the difference between the mean pupil size of each talker in the 10 seconds immediately before the start of each conversation to the mean pupil size of each talker in the last 10 seconds of each conversation.

All fitted mixed-effects models underwent diagnostic evaluations to ensure appropriate fit. Variance inflation factors were investigated to confirm that there was low collinearity across predictors (i.e., variance inflation factors < 5, determined using *check\_collinearity()* from the *performance* package in R). Residual QQ plots were inspected to verify that the distribution and dispersion of residuals did not severely deviate from the assumed distribution (using the *DHARMA* package in R for GLMMs, and *qqnorm* for LMMs). To assess the significance of the GLMM outputs, Type III Wald chi-squared tests were performed for each model (using the *car* package in R). The outputs of the LMMs were assessed using analyses of variance with the Satterthwaite approximation for degrees of freedom. To determine the direction and magnitude of any significant effects, contrast analyses of the estimated marginal means of the models were performed (using *emmeans* in R).

To analyze the resulting TRFs, care had to be taken to address serial correlation in pupil size measurements and the multiple comparisons problem. A cluster mass permutation testing approach was utilized to analyze the significance of the TRFs (Maris & Oostenveld, 2007). The procedure was carried out (in MATLAB) as follows. Each pupil response signal was randomly paired with a set of turn-taking events from a different conversation. A set of null TRFs was derived using these random pairings of temporally uncorrelated stimulus and response signals. The resulting TRFs were then analyzed for significance from zero using point-wise t-tests. The resulting t-statistics were summed over segments which were contiguously significant, and the largest resulting summed t-value was set aside. This procedure was then repeated using different randomized pairings of stimulus and response signals until there were 1000 summed t-values set aside. These values were then treated as a null distribution, and the 95<sup>th</sup> percentile was used to define a significance threshold, representing a 95% likelihood that a contiguously significant segment with a sum of t-statistics greater than this threshold was not obtained purely by chance. The TRFs based on real pairings of data were then assessed using point-wise t-tests, and contiguously significant segments with a summed t-statistic less than this threshold were discarded as non-significant.

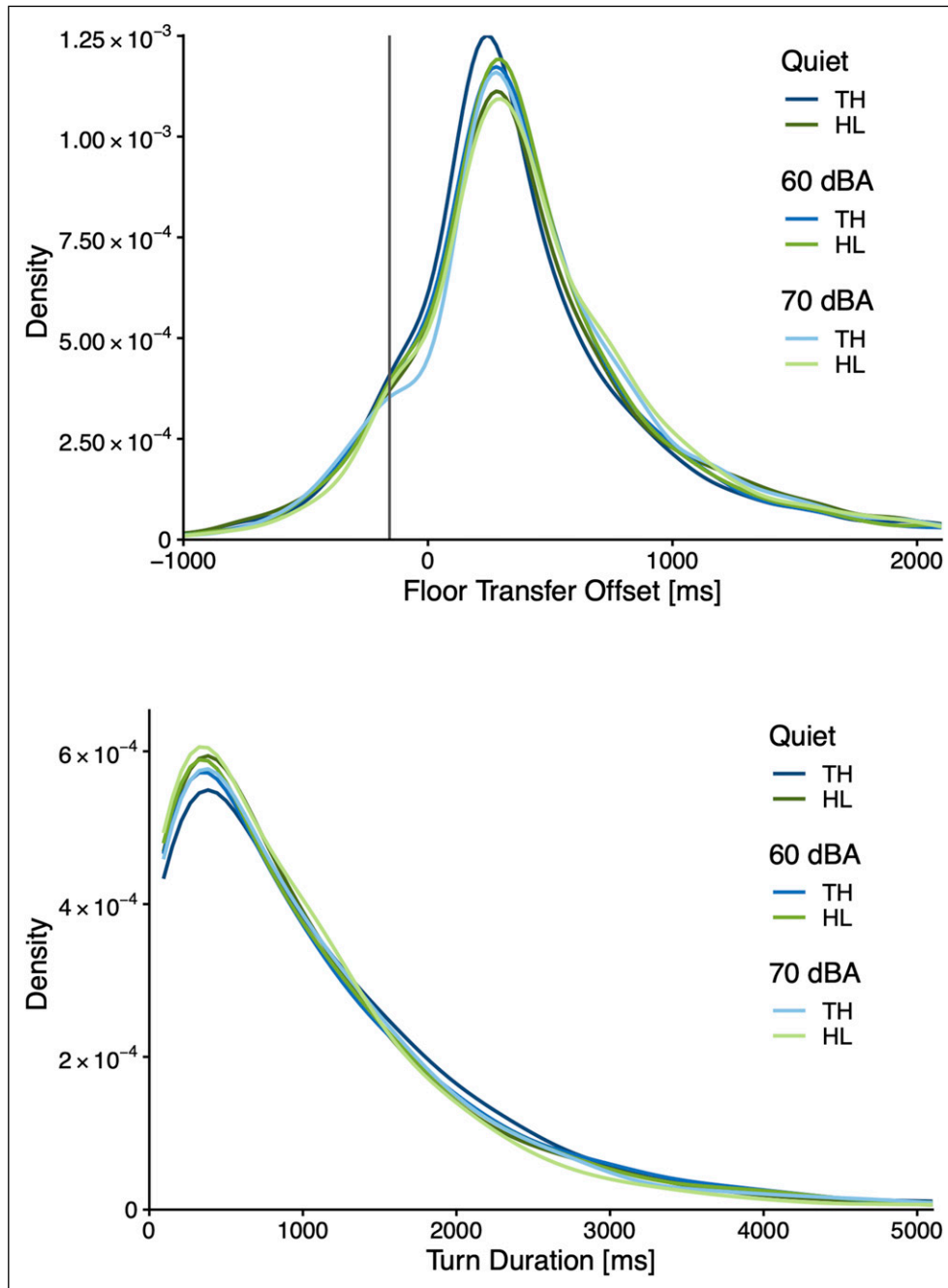
A similar procedure was also used to compare two sets of TRFs (e.g., TH versus HL, or quiet versus N70), but instead of determining the significance of a set of null derived TRFs from zero, TRFs derived from true stimulus/response pairings had their condition labels randomly shuffled, and were then compared across conditions using two-sample t-tests. In all cases, the data used construct these random pairings and null distributions were representative of the data being analyzed with the resulting threshold.

## Results

### Turn-Taking Behavior

Kernel density estimates of the distributions of FTOs and turn durations in quiet and noise for talkers with and without hearing loss are illustrated in Figure 2. The FTO distributions corresponding to the noise conditions are generally broader (i.e., have a lower peak density) and right-shifted relative to the quiet condition for TH talkers, while the noise induced differences are less clear for HL talkers. Further, the HL distributions appear broader and right-shifted relative to the TH distributions. The distributions of turn durations are generally broader for TH talkers than HL ones.

The results of the FTO GLMM are displayed in Table 1. In summary, significant main effects of background noise and the noise/hearing status interaction were observed. Pairwise contrast analysis of the estimated marginal means extracted from the



**Figure 2.** Distributions of floor-transfer offsets and turn durations in both quiet and noise (indicated by shade) for talkers with and without hearing loss (indicated by color). The vertical gray bar on the plot of FTO distributions indicates the threshold for excluding FTOs prior to fitting the GLMM

model, as shown in Table 2, revealed that TH talkers produced significantly longer FTOs in N70 relative to both quiet and N60. HL talkers produced shorter FTOs in N60 than in quiet, and longer FTOs in N70 than N60. Contrasts between hearing statuses within each noise level showed that HL talkers produced longer FTOs than TH talkers, but only in quiet. The model further revealed significant effects of the durations of the IPUs both immediately before and after the FTO, validating their inclusion as control predictors.

The results of the GLMM fitted to the distribution of turn durations are displayed in Table 3. The model revealed significant main effects of hearing status and noise. An analysis of the estimated marginal means revealed that TH talkers produced longer turns than HL talkers (1144 vs. 1043 ms,  $p = 0.044$ ). Although the Wald chi-squared tests indicated a significant effect

**Table 1.** Results of the Wald Chi-Squared Test Performed on the GLMM Fitted to the FTO Distributions

| Fixed-effect         | $\chi^2$ | df | Pr ( $>\chi^2$ ) |     |
|----------------------|----------|----|------------------|-----|
| Noise                | 43.52    | 2  | $p < 0.001$      | *** |
| Hearing Status       | 1.08     | 1  | $p = 0.29$       |     |
| Pre FT IPU Duration  | 42.48    | 1  | $p < 0.001$      | *** |
| Post FT IPU Duration | 615.62   | 1  | $p < 0.001$      | *** |
| Noise:HearingStatus  | 17.04    | 2  | $p < 0.001$      | *** |

of background noise, significance of the contrasts between background noise levels did not survive Tukey adjustment (results in Table 4), and therefore we found no significant differences in turn durations between any pairs of background noise levels.

### Conversation Level Pupillometry

The linear mixed effects model fitted to the mean pupil size in each conversation revealed a significant main effect of background noise ( $F(2, 310.07) = 15.25, p < 0.001$ ), and a borderline significant main effect of hearing loss ( $F(1, 22.44) = 4.29, p = 0.05$ ), but the interaction was not significant (see Table 5 for full results). A *post-hoc* pairwise analysis of the estimated marginal means in the three background noise conditions revealed significant differences between N70 vs. quiet (+0.089 mm), and N70 vs. N60 (+0.056 mm), but no significant difference between quiet and N60. The full results of the contrasts are summarized in Table 6. The estimated marginal means of the borderline effect of hearing loss revealed that HL talkers' pupils were trending smaller (3.59 mm TH vs. 3.33 mm HL).

The model fitted to the baseline drift parameter revealed a significant main effect of hearing status ( $F(1, 22.49) = 11.26, p = 0.003$ ), but no significant effects of noise or the interaction (see Table 7 for full results). A *post-hoc* analysis of the estimated marginal means revealed that the HL talkers exhibited a larger decrease from their baseline pupil size (0.05 mm for TH vs. 0.23 mm for HL). Figure 3 displays boxplots of the mean pupil size, and curves demonstrating pupil size drift over time.

### Pupil Responses to Turn-Taking

The computed TRFs were first grouped across all talkers and conversations and analyzed to determine intervals where the overall average TRFs were significantly different from 0. The results of this procedure are displayed in Figure 4. Significant changes in the pupil size were found to occur in the averaged TRFs corresponding to three of the four turn-taking events. Specifically, transient dilations of the pupil occur directly around self-stop, partner-start, and partner-stop.

To determine whether hearing loss is reflected in the pupil responses to turn-taking, computed TRFs were grouped according to the hearing status of the talker. Figure 5 plots the averaged TRFs of TH talkers against those of HL talkers. The responses corresponding to self-start, partner-start, and partner-stop were found to be significantly different from each other.

**Table 2.** Contrasts of the Estimated Marginal Means Extracted From the GLMM Fitted to the Distributions of Floor-Transfer Offsets. The Means Were Backtransformed to the Response Scale Prior to Computing Contrasts, and Therefore Represent the Difference, in Milliseconds, in FTOs Between the Conditions Indicated in the Contrast Column. The Presented P-values are Tukey Adjusted

| Contrast       | Estimate | Std. Err. | z-ratio | p-value     |     |
|----------------|----------|-----------|---------|-------------|-----|
| N60 - Q (TH)   | -20.5    | 13.4      | -1.53   | $p = 0.277$ |     |
| N70 - Q (TH)   | 54.0     | 14.3      | 3.78    | $p < 0.001$ | *** |
| N70 - N60 (TH) | 74.5     | 14.0      | 5.32    | $p < 0.001$ | *** |
| N60 - Q (HL)   | -76.3    | 14.5      | -5.27   | $p < 0.001$ | *** |
| N70 - Q (HL)   | -28.6    | 14.8      | -1.93   | $p = 0.129$ |     |
| N70 - N60 (HL) | 47.7     | 13.8      | 3.44    | $p = 0.002$ | **  |
| HL - TH (Q)    | 70.6     | 26.7      | 2.64    | $p = 0.008$ | **  |
| HL - TH (N60)  | 14.7     | 24.7      | 0.60    | $p = 0.551$ |     |
| HL - TH (N70)  | -12.1    | 27.0      | -0.45   | $p = 0.655$ |     |

**Table 3.** Results of the Wald Chi-Squared Test Performed on the GLMM Fitted to the Distributions of Turn-Durations

| Fixed-effect        | $\chi^2$ | df | Pr ( $>\chi^2$ ) |   |
|---------------------|----------|----|------------------|---|
| Noise               | 6.27     | 2  | $p = 0.043$      | * |
| Hearing Status      | 4.05     | 1  | $p = 0.044$      | * |
| Noise:HearingStatus | 0.61     | 2  | $p = 0.739$      |   |

To assess the effects of background noise on the pupil responses to turn-taking, TRFs computed from conversations taking place in each background noise condition were compared to all others. These comparisons were done separately for TH and HL talkers. Here we present only the comparisons between quiet and N70 for conciseness. Figures illustrating the comparisons between both conditions and N60 are provided in the Appendix (see [Figure A1](#), [Figure A2](#), [Figure A3](#), & [Figure A4](#)). In general, N60 did not differ from Quiet for either TH or HL talkers, while the responses in N70 differ similarly from N60 as they do from quiet.

[Figure 6](#) demonstrates the pupil responses to turn-taking of TH talkers in quiet versus N70, and [Figure 7](#) performs the same comparison for HL talkers. While [Figure 6](#) demonstrates that different responses are observed in quiet versus N70 around all four turn-taking events for TH talkers, [Figure 7](#) illustrates that significant differences are observed only around partner-stop for HL talkers.

## Discussion

### *Behavioral Markers of Turn-Taking Related Effort*

The significant increase in FTO duration under the high background noise, as observed for TH talkers, aligns with numerous previous studies ([Ellag et al., 2025](#); [Masters & MacDonald, 2025](#); [Petersen et al., 2022](#); [Sørensen et al., 2021, 2024](#)). However, the N60 condition reveals a non-monotonic effect of background noise in HL talkers, as FTOs were longer in N70 than N60, but shorter in N60 than in quiet. Therefore, HL talkers showed no significant difference in FTO durations between quiet and N70. These findings differ from [Sørensen et al. \(2024\)](#), which compared FTOs produced by younger TH talkers conversing with older HL talkers. In that study, HL talkers produced similar FTOs to their younger TH counterparts, and both groups' FTOs were similarly impacted by the presence of noise. Here, we observed similar FTO durations between TH and HL talkers in both background noise conditions, but the FTOs produced by the HL talkers were longer than TH in quiet.

Several factors may contribute to these differences. First, in the current study, both the HL and TH talkers were older, which could change the social dynamic and therefore result in different adaptive behavior. In [Sørensen et al. \(2024\)](#), older HL talkers conversed with younger TH talkers, which may have introduced social pressure to maintain a faster conversational pace, therefore reducing FTO durations in quiet. In the present study, the closer alignment in age of the talkers could have created a more relaxed social dynamic, resulting in HL talkers feeling more comfortable using the quiet conversation to rest and recover cognitive resources, and therefore taking longer pauses between turns, increasing their FTOs. This resting process could then have been disrupted once noise was introduced, resulting in the decrease in FTOs observed in N60. Additionally, in [Sørensen et al. \(2024\)](#) the talkers were situated in separate booths and could not see each other. Thus, some differences might be explained by access to non-acoustic forms of communication which could affect speech production, for example by replacing verbal responses with nodding or other gestures. Further, [Sørensen et al. \(2024\)](#) recruited only HL talkers with mild-to-moderate hearing loss, whereas in the present study ~38% of HL participants had moderate or severe hearing loss. Therefore, it is also possible that the higher difficulty introduced by the more substantial degrees of hearing loss could have partially driven this

**Table 4.** Pairwise Contrasts, Over Background Noise Level, of the Estimated Marginal Means Extracted From the GLMM Fitted to the Distribution of Turn-Durations. The Presented P-values are Tukey Adjusted

| Contrast  | Estimate | Std. Err. | z-ratio | p-value     |
|-----------|----------|-----------|---------|-------------|
| N60 - Q   | -4.78    | 19.9      | -0.24   | $p = 0.969$ |
| N70 - Q   | -44.53   | 19.7      | -2.26   | $p = 0.061$ |
| N70 - N60 | -39.75   | 19.4      | -2.05   | $p = 0.101$ |

**Table 5.** Results of the Analysis of Variance Performed on the Linear Mixed Effects Model Fitted to the Mean Pupil Size in Each Conversation

| Fixed-effect        | F     | df          | p-value         |
|---------------------|-------|-------------|-----------------|
| Noise               | 15.25 | (2, 310.07) | $p < 0.001$ *** |
| Hearing Status      | 4.29  | (1, 22.44)  | $p = 0.050$ .   |
| Noise:HearingStatus | 0.17  | (2, 310.07) | $p = 0.841$     |

effect. HL participants with more loss could exhibit a shift in communication strategy when things become too difficult in noise. For example, they may utilize more non-verbal communication or rely on shorter responses to reduce speech planning effort, as suggested by the observation that HL talkers produced shorter turns on average than TH ones.

### *Correspondence of conversation-level metrics and studies of listening effort*

We observed that the pupil size is, on average, significantly larger in noisy conversations and we also found a borderline effect suggesting that pupil size is trending smaller in HL talkers compared to TH ones. The effect of noise is expected as, effort aside, the presentation of a louder auditory stimulus will generally induce a larger dilation of the pupil than a quieter one (Zekveld et al., 2018). However, we also expect that increased effort contributed to the larger pupil size in noise, as comprehending speech in noise will be more difficult than in quiet, and we expect that this increased load will also increase the difficulty of the concurrent planning of upcoming speech. Although no studies, to our knowledge, have applied pupillometry to conversations involving HL talkers, the smaller pupil size observed aligns with previous findings from degraded speech perception tests (Wang et al., 2018; Zekveld et al., 2023). Of note, though, is that the previous speech perception studies only consisted of listening tasks. In the present study, we also expect that increased effort related to speech preparation, planning, and production would contribute to our findings. Zekveld et al. (2023) also found that HL talkers exhibited a steeper drift from baseline across an experimental session, a finding which aligns with our evaluation of drift across individual conversations. One suggested explanation given for this experimental drift was fatigue, given that the HL talkers also exhibited a stronger decrease in performance across trials (Zekveld et al., 2023). We suspect that increased fatigue may also explain the steeper drift we observe here. Given that the FTO distributions of HL talkers in quiet somewhat resemble the FTO distributions of TH talkers in noise, it is possible that the quiet conversations had a higher baseline level of difficulty for HL talkers than TH ones. Therefore, while quiet conversations may have been less effortful for TH talkers and could have served as rest, to some degree, the same may not have been true for HL talkers. Further, HL talkers exhibiting higher fatigue could lead to an adjustment of communication strategy by the talkers, which may also partially explain the differences in the effect of background noise on turn-taking behavior by HL talkers discussed previously.

### *Consistent Pupil Responses to Turn-Taking Are Observed*

The pupil was found to respond consistently to three of the four turn-taking events across talkers, hearing status, and background noise conditions. Specifically, dilations of the pupil occurred immediately around self-stop, partner-start, and partner-stop. Some of these dilations may indicate preparatory and reactionary direction of listening effort, for example switching attention to prepare to listen around self-stop, and just before partner-start, and beginning to listen just after partner-start. Further, the significant response observed around partner-stop could be indicative of effort directed to speech planning. Although it may seem more intuitive that speech planning effort would be observable as an anticipatory component in the pupil response to self-start, previous studies have shown that people begin planning their next turn as soon as they are able

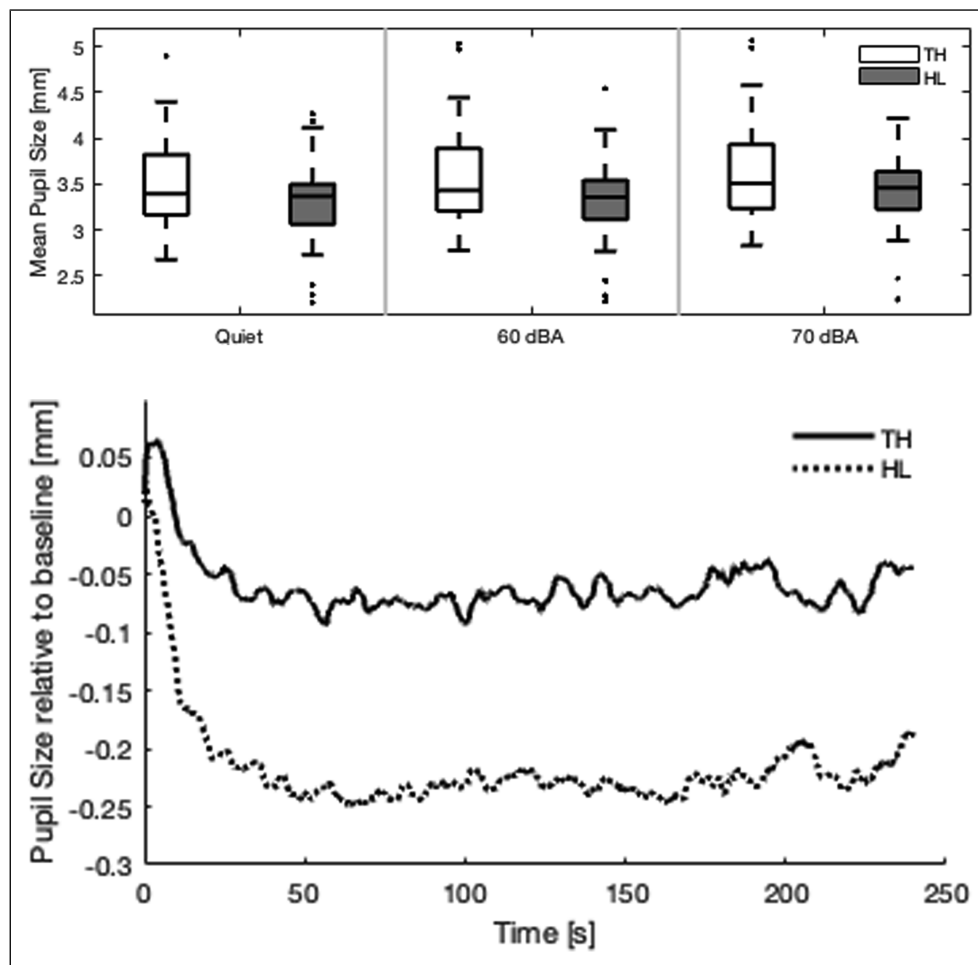
**Table 6.** Pairwise Contrasts, Over Background Noise Level, of the Estimated Marginal Means Extracted From the LMM Fitted to the Mean Pupil Size in Each Conversation. The Presented P-values are Tukey Adjusted

| Contrast  | Estimate | Std. Err. | df  | t-ratio | p-value         |
|-----------|----------|-----------|-----|---------|-----------------|
| N60 - Q   | 0.03     | 0.021     | 310 | 1.99    | $p = 0.118$     |
| N70 - Q   | 0.09     | 0.019     | 310 | 5.46    | $p < 0.001$ *** |
| N70 - N60 | 0.06     | 0.018     | 310 | 3.45    | $p = 0.002$ *** |

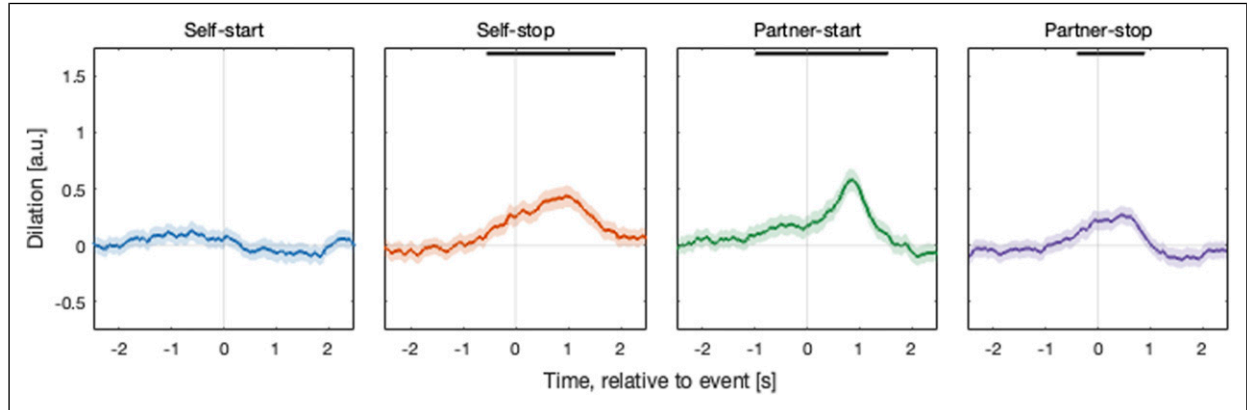
**Table 7.** Results of the Analysis of Variance Performed on the Linear Mixed Effects Model Fitted to the Baseline Drift of the Pupil Over Each Conversation

| Fixed-effect        | F     | df          | p-value     |    |
|---------------------|-------|-------------|-------------|----|
| Noise               | 1.68  | (2, 312.45) | $p = 0.189$ |    |
| Hearing Status      | 11.26 | (1, 22.49)  | $p = 0.003$ | ** |
| Noise:HearingStatus | 2.47  | (2, 312.56) | $p = 0.086$ |    |

(Bögels, 2020; Bögels et al., 2018), and therefore this process may be more closely aligned in time with the end of the partner's turn, around which a significant dilation of the pupil was observed. Prior research on pupil responses around speech planning and production, in non-conversational settings, tends to evaluate responses in a manner which is time-locked to the point at which planning can begin (e.g., when talkers are presented with an image to describe), rather than to evaluate responses prior to the point at which production begins (Barthel & Sauppe, 2019; Sauppe, 2017). During conversation, the point at which planning begins can be highly variable and is dependent on the content of a partner's turn. Therefore, it is possible that the timing of planning may be more tightly related to the end of a partner's turn rather than to the start of one's own. Thus, it is not clear whether one would expect a consistent pupil response before self-start. Even though planning must precede production, it is possible this planning process appears in the TRFs as a consistent response around partner-stop, rather than as a response consistently occurring synchronously ahead of self-start, given that both events occur before the onset of production.



**Figure 3.** Boxplots of mean pupil size and plots indicating pupil size trends over the course of a conversation. Boxplots are grouped by whether the conversation took place in the presence or absence of background noise, and shading indicators either typical hearing (no shading) or hearing loss (dark shading). Pupil trends over time are visualized by averaging across all conversations for typical hearing talkers (solid line) and talkers with hearing loss (dotted line)

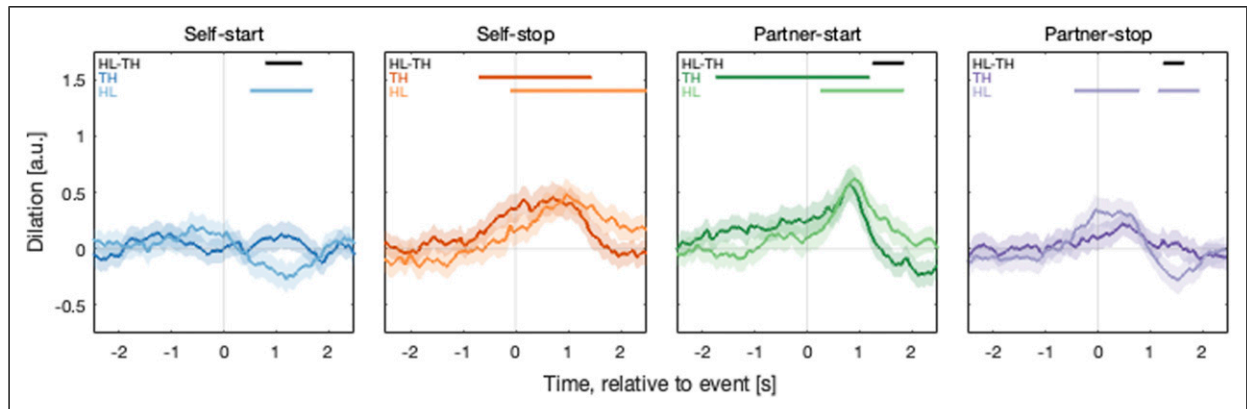


**Figure 4.** Pupillary temporal response functions pooled across all talkers and conversations, for each of the four turn-taking events. The shaded regions indicate 95% confidence intervals. The black bars indicate regions where the set of TRFs have a mean significantly different from zero

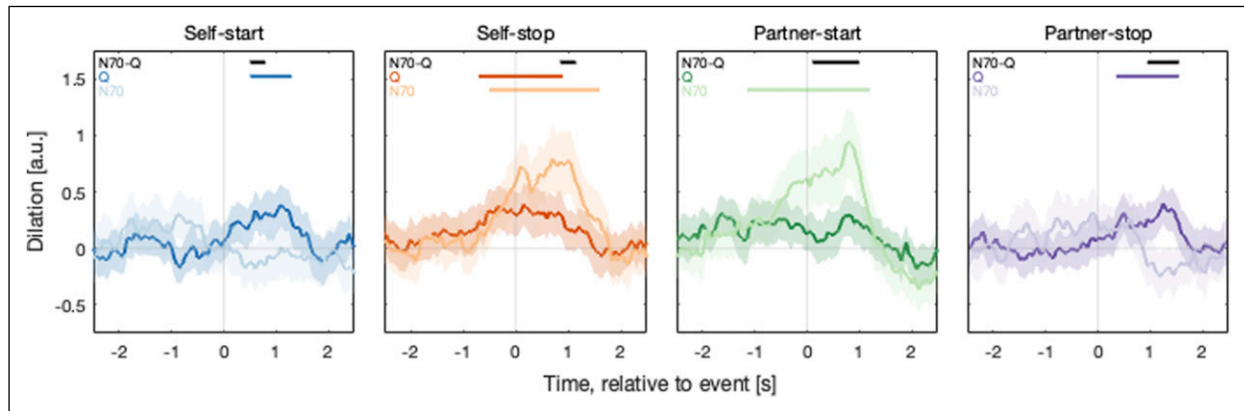
The responses observed in [Figure 4](#) generally align in morphology and direction with the findings of [Masters et al. \(2025\)](#), indicating that similar pupil responses to turn-taking can be observed across different experiments. Of note, though, is that both studies recruited participants with similar demographics (e.g., age, first-language). Thus, it remains unclear if the effects observed here would generalize to other populations. However, the groups recruited for these studies align in age with the demographic which most commonly receives evaluation and treatment for hearing loss. The two experiments also both utilized the same conversational spot-the-difference task ([Baker & Hazan, 2011](#)), and therefore it is worth exploring in future studies if similarly consistent responses can be observed in more free-form conversations, which may be representative of a more naturalistic communication environment.

### *Hearing Loss and Background Noise Modulate Pupil Responses*

By aggregating pupil responses to turn-taking across all conversations for TH and HL talkers separately, as shown in [Figure 5](#), significant effects of hearing status were observed on the responses to all turn-taking events, except self-stop. Around 1 second after self-start, HL talkers exhibit a constriction of the pupil resulting in a significant difference from their TH counterparts. Similarly, a constriction of the pupil is observed in HL talkers, but not TH ones, after partner-stop. One possible explanation for these responses is that HL talkers experience a greater release from effort after they start speaking. This could result from listening being relatively more effortful for HL talkers than speaking is. This seems plausible, seeing as HL talkers exhibited a larger dilation of the pupil than TH talkers around 1 second after the partner-start event. Although one may expect



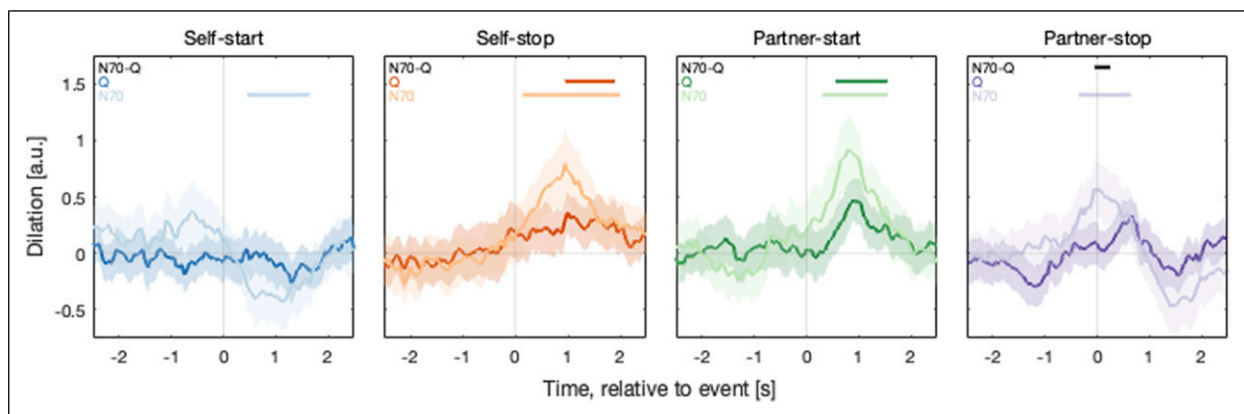
**Figure 5.** Averaged TRFs across all conversations of TH talkers (darker curve) and HL talkers (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two groups are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero



**Figure 6.** Averaged TRFs of TH talkers across conversations taking place in quiet (darker curve) and 70 dBA background noise (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two conditions are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero

this effect to be visible through a larger peak dilation of the pupil, as well, it is important to note that pupil size is standardized within talkers. Therefore, the largest dilations exhibited by each group will have similar magnitudes.

Taken in tandem, these findings could be explained by differences in capacity for concurrent listening and speech planning. If listening demands a higher resource allocation in HL talkers, as would be expected, then they will have less resources available to simultaneously plan their next segment of speech. Thus, the speech planning process would be delayed in HL talkers, as is also suggested by our findings on the FTOs in quiet. This would then require sustained effort during listening, resulting in the differences observed in the partner-start response. Further, once speaking begins and this highly demanding concurrent processing is no longer required the HL talkers would experience a greater release, due to their increased effort just prior. This possibility is supported by a previous study which found that increased listening demand impairs turn-end prediction and delays response planning (Corps et al., 2025). However, it is also possible that the sustained dilation of HL talkers observed around partner-start is driven by differences in turn duration. Given that TH talkers were found to produce longer turns than their HL counterparts, the HL talkers were listening for longer, on average, which could partially contribute to the sustained dilation of the pupil after partner-start. A *post-hoc* evaluation of the estimated marginal means extracted from the GLMM fit to turn durations revealed that TH talkers produced turns that were ~100 ms longer on average. Yet, the sustained region of dilation after partner-start occurs for ~500 ms. Therefore, although differences in turn durations may partially explain the sustained response, they do not fully account for the result. Further, if listening demand is lower in TH talkers, they may be able to begin planning their upcoming turn earlier during their partner's turn. Therefore, the



**Figure 7.** Averaged TRFs of HL talkers across conversations taking place in quiet (darker curve) and 70 dBA background noise (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two conditions are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero

point at which speech planning starts to occur could become more variable and less synchronous with the end of the partner's turn and thus induce less consistently timed evoked pupil responses. This variability could also explain why the response of TH talkers to partner-stop was never observed to be significantly different from zero.

The findings presented in [Figures 6 and 7](#) demonstrate the effects of background noise on pupil responses for TH and HL talkers. TH talkers exhibit significantly different pupil responses between quiet and N70 around all four turn-taking events. The responses of HL talkers only differ between quiet and N70 around partner-stop. Qualitatively, the responses of HL talkers in both quiet and N70 are morphologically similar to the responses of TH talkers in N70 (e.g., the dilation occurring after partner-start). These observations corroborate the idea that conversing in quiet may have a higher baseline difficulty for HL talkers than TH ones. Although previous studies have examined the pupil responses to speech-in-noise versus speech-in-quiet and found no differences between TH and HL listeners in quiet ([Kramer et al., 2016](#); [Zekveld et al., 2023](#)), the patterns observed here suggest that, although listening in quiet may not be measurably more effortful for HL participants, conversing in quiet can be. This finding, taken alongside the effects observed on the FTO distributions, reveals that quiet conversations impose greater difficulty on HL talkers than TH ones.

Despite the lack of significant differences between quiet and N70 in the pupil responses to turn-taking for the HL talkers, the conversation-level analysis revealed that background noise did significantly increase pupil size. In combination with the morphological observations of the TRFs, it is possible that fatigue could partially explain the lack of noise effect on the TRFs, as fatigue has been shown to result in smaller evoked pupil responses ([Wang et al., 2018](#)), and therefore may lead to saturation of turn-evoked responses in noise for HL talkers.

## Limitations

Given the relatively open nature of both the experimental setup and of conversation, it is difficult to definitively state whether the responses observed here truly reflect differences in cognitive effort. For example, the gaze of talkers was not restricted, and thus accommodation or local differences in luminance could be partially driving the pupil responses. One advantage of the TRF approach, though, is that to affect the pupil responses to turn-taking, such factors would also need to vary consistently around these events throughout the conversations. Further, although we included covariates to capture accommodation related pupil responses to account for the possibility that talkers adjust their gaze in some manner systematically related to turn-taking, if an accommodation effect were to dominate the pupil response over a given interval, then the TRF method may not be capable of appropriately disentangling gaze and turn-taking related responses. However, to diminish this concern, a post-hoc analysis was conducted based on the gaze depth signal, revealing that talkers spent, on average, less than 5% of the duration of each conversation looking at their partner in all conditions. Despite the potential alternate sources for these pupil dilation responses, the findings here still demonstrate that both background noise and hearing loss modulate the factors which drive these pupil responses, suggesting that they are, indeed, indicative of difficulty. Another related limitation around the TRF analysis is the inability to include all desired factors as predictors or covariates. For example, a *post-hoc* approach had to be taken to investigate the possibility of differences in turn durations contributing to the sustained pupil responses of HL talkers, as opposed to the direct inclusion of such a factor in the TRF model itself. Although there are other modelling approaches which can include such effects (e.g., GAMMs), those approaches do not possess the deconvolutional nature of TRFs, and thus corruption of responses from nearby events becomes a concern.

A limitation of the secondary analysis conducted here is the presence of an unanalyzed experimental factor: the 3 noise reduction settings for the HL talkers. These settings likely introduced asymmetries in the difficulty experienced by the TH and HL talkers. However, despite this potential reduction in listening difficulty for HL talkers in some conversations, we still observed significant differences in their pupil responses and turn-taking behavior. Had these settings not been in place, the conversations may have been more difficult, on average, for HL talkers and therefore the effects of hearing status might have been more pronounced.

Another potential limitation of the present analysis is that our detection of turns is based only on correspondence of voice activity detection across talkers. Thus, we did not consider what was being said, but rather only when speech was occurring. It is possible that turns consisting of more brief responses (e.g., 'mhm') would require less effort in terms of planning/production and listening than other more complicated turns. An interesting future direction of research would be the development of an automated toolset which could classify and cluster turns based on their content and consider that information in such turn-taking analysis.

## Conclusions

We showed that background noise and hearing loss are reflected in pupil responses during conversations. Further, we demonstrated that although there is correspondence between pupil responses and differences in turn-taking behavior, the

behavioral adaptations do not fully explain the pupil responses. This supports the notion that both are indicative of some higher-level processing differences resulting from increased difficulty. Our findings suggest that conversing in quiet is considerably effortful for HL talkers, relative to their TH counterparts. Further work is needed to confidently regard the effects observed here as being related to cognitive effort, though, as they could be reflective of behavioral adaptations, e.g. adjustments of gaze, which induce responses of the pupil.

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### Ethical Considerations

The secondary analysis of data in the present study was approved by the University of Waterloo's Research Ethics Committee (REB No. 45442).

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### Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

### Data Availability Statement

Deidentified data that support the findings of this study are publicly available on Borealis ([Masters et al., 2026](#)).

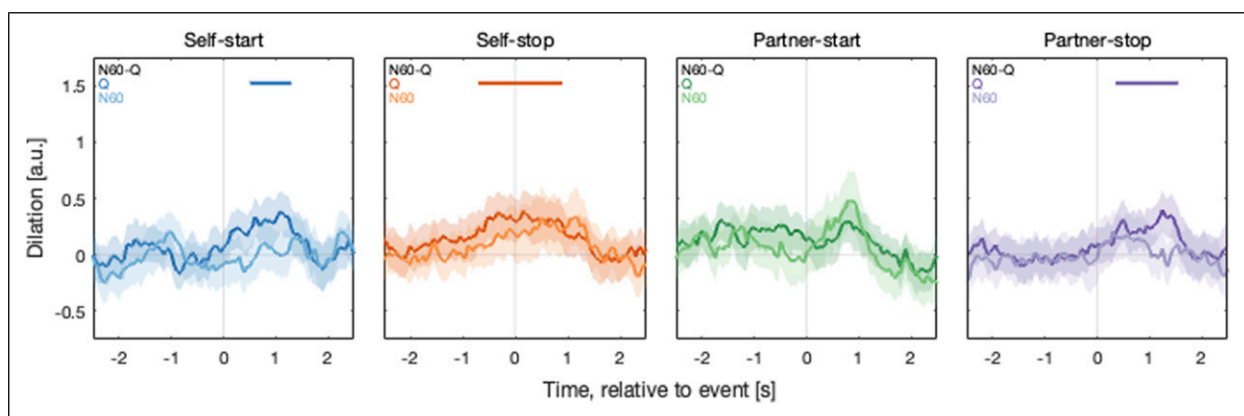
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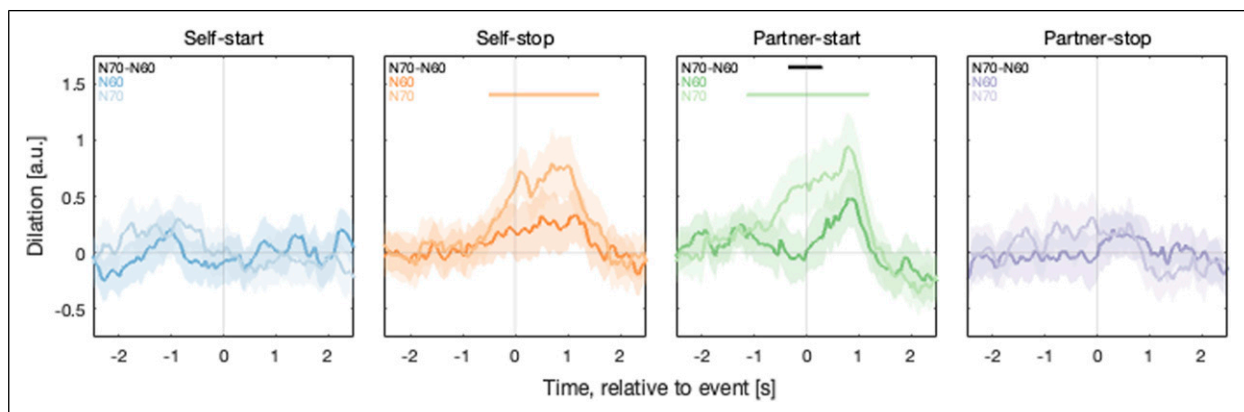
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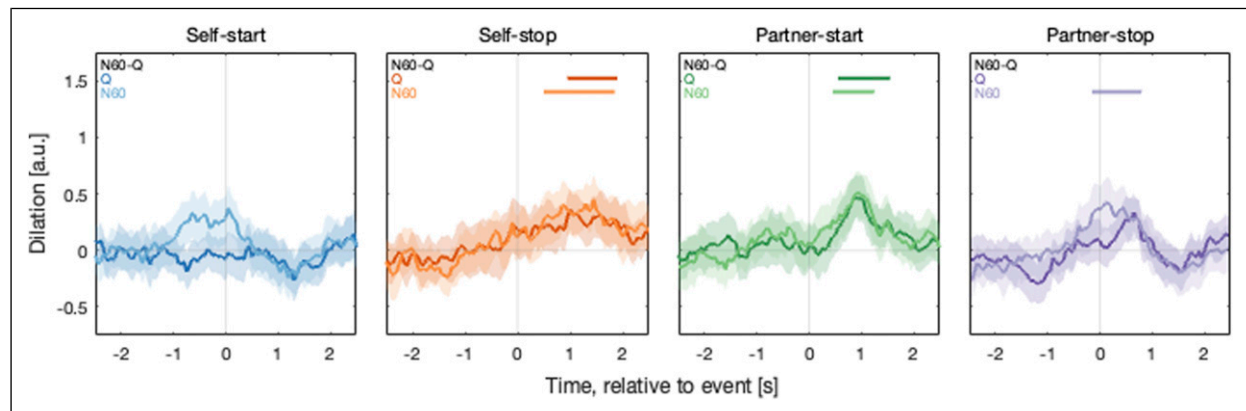
## Appendix



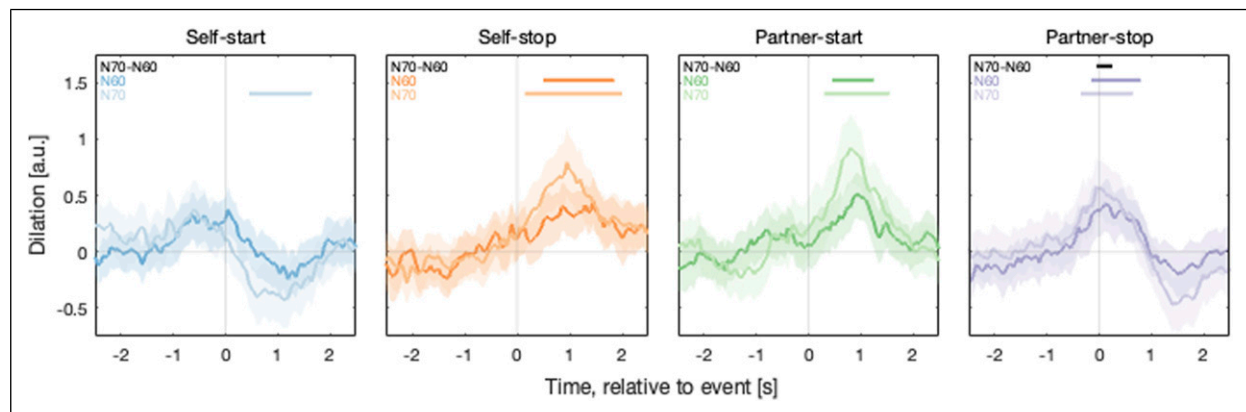
**Figure A1.** Averaged TRFs of TH talkers across conversations taking place in quiet (darker curve) and 60 dBA background noise (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two conditions are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero.



**Figure A2.** Averaged TRFs of TH talkers across conversations taking place in 60 dBA (darker curve) and 70 dBA background noise (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two conditions are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero.



**Figure A3.** Averaged TRFs of HL talkers across conversations taking place in quiet (darker curve) and 60 dBA background noise (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two conditions are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero.



**Figure A4.** Averaged TRFs of HL talkers across conversations taking place in 60 dBA (darker curve) and 70 dBA background noise (lighter curve). Shaded regions indicated 95% confidence intervals of each group. Black bars indicate regions where the means of the two conditions are significantly different from each other. The color matched bars indicate regions where each of the curves are significantly different from zero.